CS 4400: Artificial Intelligence

Intelligent Agents

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2. Task environments
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3. Agent architecture

Discussion

- Godel's theorem assures us that humans will always be superior to machines.
- A robot/agent can never be aware of itself (be self-conscious).

1/3. Agent

- An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators
- **Human agent:**

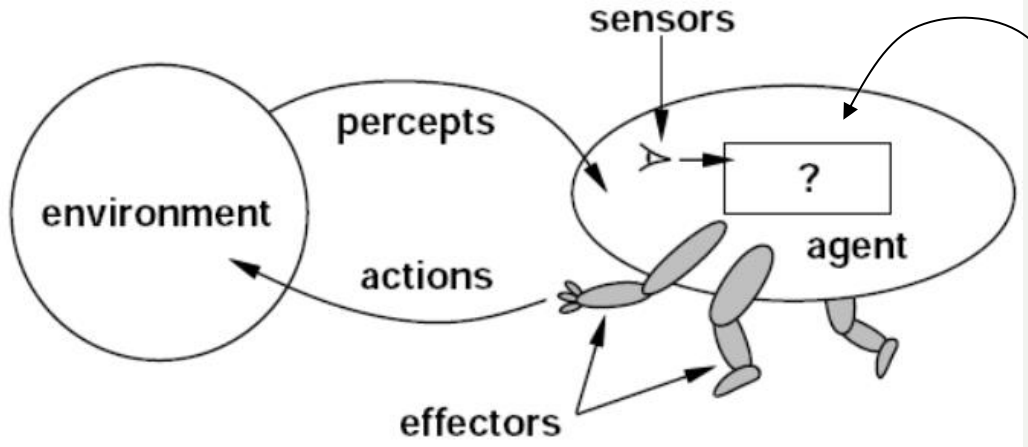
  eyes, ears, and other organs for sensors;

  hands, legs, mouth, and other body parts for actuators
- **Robotic agent:**

  cameras and infrared range finders for sensors;

  various motors for actuators

1/3. Agent and environments



- Perception (sensors)
- Reasoning / cognition
- Action (actuators)

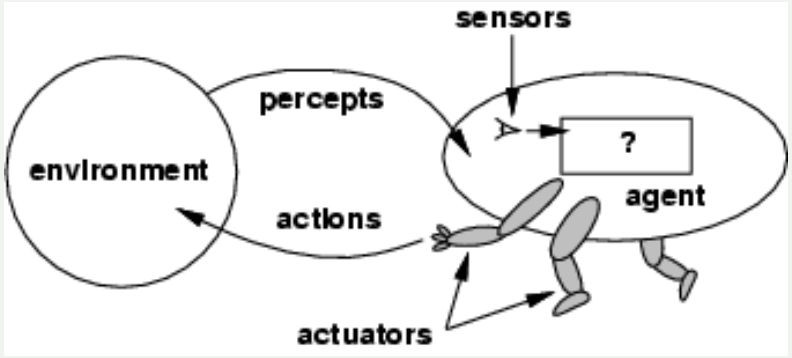
Percept: perceptual inputs at any given instant.

Percept sequence: complete history of everything the agent has ever perceived.

Agent function (behavior):

a percept sequence $\overset{\text{reasoning}}{\mapsto}$ an action

1/3. Agent and environments



- The agent function maps from percept histories to actions:

$$[f: \mathcal{P}^* \rightarrow \mathcal{A}]$$

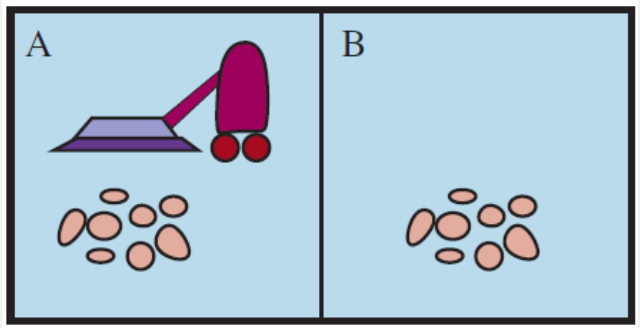
- The agent program runs on the physical architecture to produce *f*
- agent = architecture + program

Vacuum world

Two locations (A & B)

Vacuum agent perceives the location and the presence of dirt

Vacuum agent can move left, right, suck, nothing



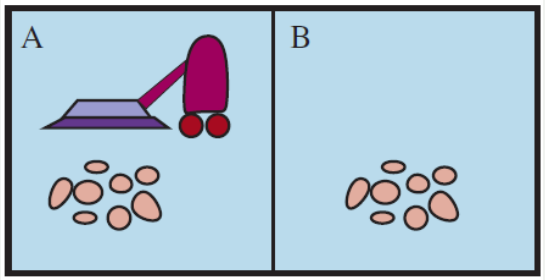
- *Environment*: squares A & B
- *Percepts*: [A, *Dirty*]

square the vacuum cleaner is in

state of the square

- *Actions*: *left*, *right*, *suck*, *nothing*

Agent function



Percept sequence	Action
$[A, Clean]$	$Right$
$[A, Dirty]$	$Suck$
$[B, Clean]$	$Left$
$[B, Dirty]$	$Suck$
$[A, Clean], [A, Clean]$	$Right$
$[A, Clean], [A, Dirty]$	$Suck$
⋮	⋮

An agent is fully characterized by the agent function: *percept sequence* → *action*

Evaluating the basic agent

Need a performance measure to evaluate how desirable an agent's behavior is

- Maximize amount of dirt picked up?
- Maximize ratio of dirt to energy expended?
- Maximize ratio of dirt to combination of energy and time?

Wrong performance measures lead to rational but unwanted outcome

- Vacuum: Maximize amount of dirt → ?
- The Matrix/Skynet: End pain and bring peace to the world → ?

Rationality

- Rational Agent: For each possible percept sequence, a rational agent should select an action that is *expected* to maximize its performance measure, based on the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

- What is rational depends on four things:

  - performance measure An objective criterion for success of an agent's behavior
  - prior knowledge of the environment
  - performable actions by the agent
  - perceptual sequence to date

  A rational agent should select an action expected to maximize its performance measure.

Omniscience vs Rationality

- An *omniscient agent* knows the actual outcome of its actions.

Impossible in reality!

- Rationality maximizes the expected performance.
  - Learn as much as it perceives.
  - Does not require omniscience.
- Perfection maximizes actual performance.

Rationality ≠ omniscience ≠ perfection

Rationality

The definition of rationality implies that a rational agent should be able to

- explore the world to gather information
- learn from what it perceives
- autonomous rather than follow what it's designed to do

2/3. Task environments

Task environment

- is the “problem” we would like to solve
- involves both the agent and the environments

Task environments are best described by PEAS description

- P: Performance measure
- E: Environment
- A: Actuators (= effectors)
- S: Sensors

PEAS for vacuum problem

Performance measure

- Maximize ratio of dirt to energy/time usage

Environment

- Two squares A&B, with/without dirt

Actuators

- Wheels, vacuum

Sensors

- Location, dirt sensor

PEAS for automated taxi

Performance measure

Safe, fast, legal, economical, and comfortable trip

Environment

Road, traffic, pedestrian, passenger

Actuators

Steering, accelerator, brake, signal, horn, display

Sensors

Cameras, rangefinder, speedometer, GPS, odometer, accelerometer, etc.

PEAS for Mars Rovers

Performance measure

Terrain explored and reported, samples gathered and analyzed

Environment

Vehicle, lander, Mars

Actuators

Wheels/legs, sample collection device, analysis device, radio transmitter

Sensors

All things in automated taxi and more

PEAS for Medical diagnosis system

Performance measure

- Healthy patient, minimize costs, lawsuits

Environment

- Patient, hospital, staff

Actuators

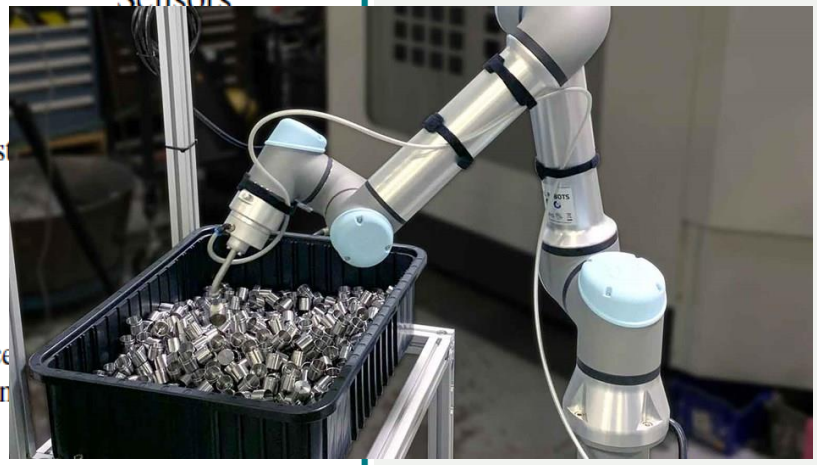
- Screen display (questions, tests, diagnoses, treatment, referrals)

Sensors

- Keyboard (entry of symptoms, findings, patient's answers)

PEAS for Other Agents

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tes diagnoses, treatments	
Satellite image analysis system	Correct categorization of objects, terrain	Orbiting satellite, downlink, weather	Display of sc categorizatio	
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, tactile and joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, raw materials, operators	Valves, pumps, heaters, stirrers, displays	Temperature, pressure, flow, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, feedback, speech	Keyboard entry, voice



Properties of task environments

PEAS describes the basics components of a task environment

To design the agent, the properties of the task environment should be further analyzed

- Fully, partially, or not observable
- Deterministic or stochastic
- Single or multi-agent
- Episodic or sequential
- Static or dynamic
- Discrete or continuous

Properties of task environments

Fully observable (vs. partially observable): An agent's sensors give the agent access to the complete state of the environment at each point in time.

Observability

Can the agent get all *relevant* information from the environment using its sensors?

Fully observable

- Chess: Positions of all the playing pieces
- AI enemy in first-person shooter video game

Partially observable

- Poker: Only the cards in your hand
- “You” in first-person shooter game

Unobservable

- No sensor at all

Properties of task environments

Deterministic (vs. stochastic): The next state of the environment is completely determined by the current state and the action executed by the agent. (If the environment is deterministic except for the actions of other agents, then the environment is strategic)

Deterministic vs Stochastic

Is the next state of the world *determined completely* by the current state of the world and the action?

Vacuum world

Deterministic: suction and motor wheels always work

Stochastic: unreliable suction and/or motor wheels

Chess: deterministic, except for the actions of other agents

Automated taxi: stochastic

Single agent (vs. multi-agent): An agent operating by itself in an environment. Does the other agent interfere with my performance measure?

Single vs Multi-agent

Competitive, cooperative, or mixed

Properties of task environments

Episodic (vs. sequential): if the agent's experience is divided into atomic episodes, among which one does not depend on the actions taken in previous ones.

- Episodic vs. Sequential
  - Episodic: current action affects only the current episode
  - Sequential: future episodes as well (e.g., chess, or taxi driving)
- Static vs. Dynamic
  - Static: time halts while agent makes a decision
    - Turn-based games (e.g., Monopoly)
  - Dynamic: world changes on while agent decides
    - Pac-man
  - Semi-dynamic: world stands still, but decision time has an effect on performance score
    - Tournament-level chess

Properties of task environments

Discrete (vs. continuous): A limited number of distinct, clearly defined percepts and actions.

Discrete vs. Continuous

- Is time handled in chunks or continuously?
- Is a state … ?
- Is an action … ?
- Examples: chess vs automated taxi

Known vs. Unknown

- Known: agent knows how its action affects the state of the world
- Unknown environments (Part 3 Reinforcement learning)

Examples of task environments

Task Environment	Observable	Deterministic	Episodic	Static	Discrete	Agents
Robot soccer						
Internet book-shopping						
Autonomous Mars rover						

Internet book-shopping: buying a used AI textbook online

- P: quality/price
- E: used book sales websites (E-bay, Amazon, Craigslist, etc)
- A: clicking links or buttons
- S: web-browser

Examples

Task Environment	Observable	Deterministic	Episodic	Static	Discrete	Agents
Robot soccer	Partially	Stochastic	Sequential	Dynamic	Continuous	Multi
Internet book-shopping						
Autonomous Mars rover						

Examples

Task Environment	Observable	Deterministic	Episodic	Static	Discrete	Agents
Robot soccer	Partially	Stochastic	Sequential	Dynamic	Continuous	Multi
Internet book-shopping	Partially	Deterministic*	Sequential	Static*	Discrete	Single
Autonomous Mars rover						

Internet book-shopping

- Sequential? Episodic?
- Static? Dynamic?
- Single-agent? Multi-agent?

No absolutely true answer; depends on how to formulate the task

Examples

Task Environment	Observable	Deterministic	Episodic	Static	Discrete	Agents
Robot soccer	Partially	Stochastic	Sequential	Dynamic	Continuous	Multi
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Real world problems:

partially observable, stochastic, sequential, dynamic, continuous, multi-agent

Properties of the task environment largely determines the agent design

task environm.	observable	determ./ stochastic	episodic/ sequential	static/ dynamic	discrete/ continuous	agents
crossword puzzle	fully	determ.	sequential	static	discrete	single
chess with clock	fully	strategic	sequential	semi	discrete	multi
poker						
back gammon						
taxi driving	partial	stochastic	sequential	dynamic	continuous	multi
medical diagnosis	partial	stochastic	sequential	dynamic	continuous	single
image analysis	fully	determ.	episodic	semi	continuous	single
partpicking robot	partial	stochastic	episodic	dynamic	continuous	single
refinery controller	partial	stochastic	sequential	dynamic	continuous	single
interact. Eng. tutor	partial	stochastic	sequential	dynamic	discrete	multi

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3/3. Agent architecture

Five different designs of agents

- Reflex agent
- Model-based reflex agent
- Goal-based agent
- Utility-based agent
- Learning agent

Reflex agent

Simplest kind of agents

Choose action based on the current percept, ignoring the percept history

Biologic organisms: sub-agents of a bigger agent

- Closing eyes when something approaches eyes
- Move the hand away when it touches something hot

Reflex agent

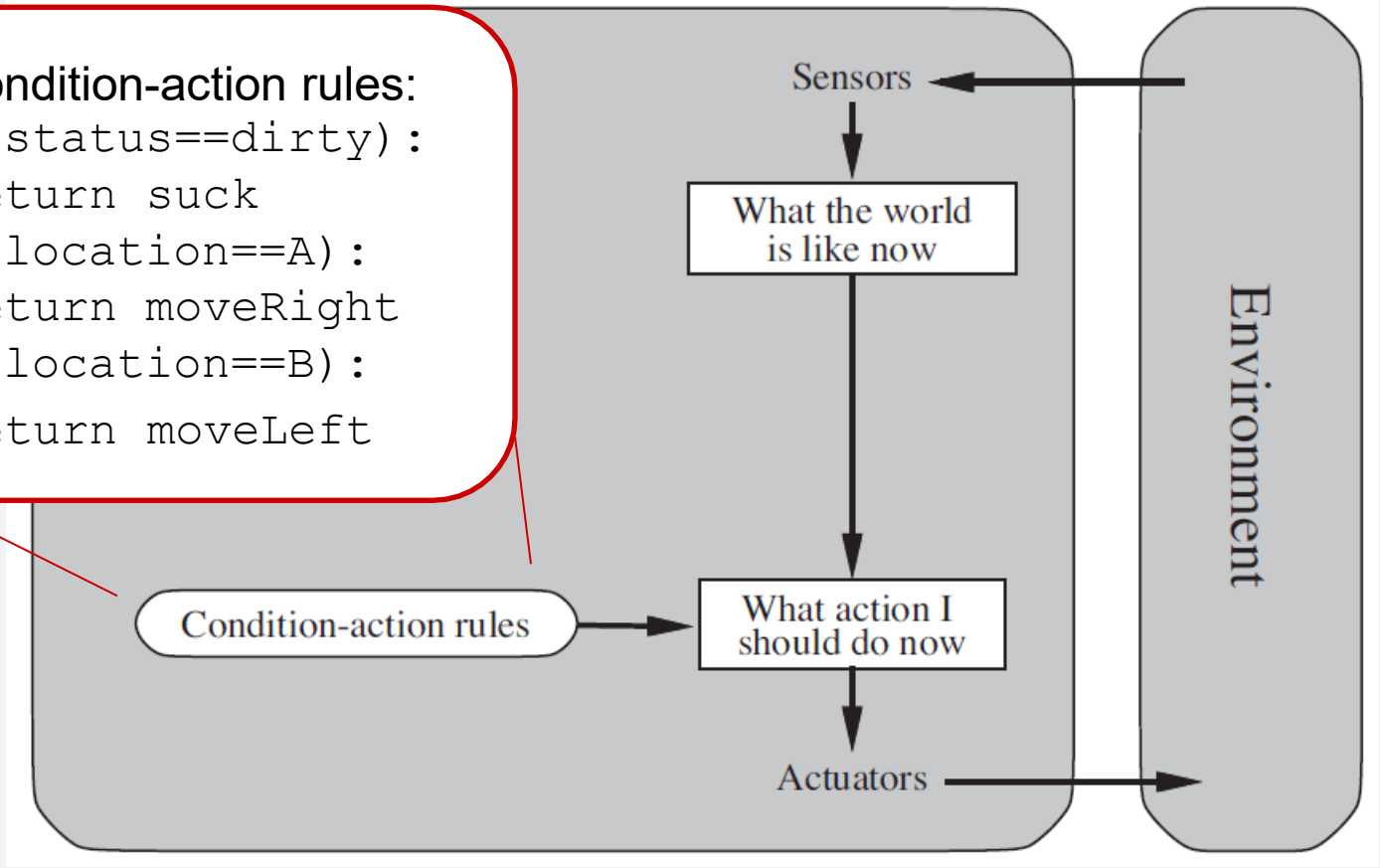


NO MEMORY
Fails if environment is partially observable

- Implemented through *condition-action* rule.

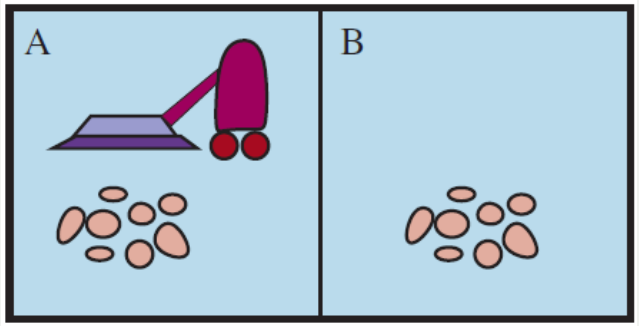
Condition-action rules:

```
if (status==dirty):
  return suck
if (location==A):
  return moveRight
if (location==B):
  return moveLeft
```



Rectangles: agent's current internal state
Ovals: background information used in the process.

Vacuum-Cleaner World (Revisited)



if *status* == *Dirty* then return *Suck*
else if *location* == *A* then return *Right*
else if *location* == *B* then return *Left*

Simple Reflex Agent

function SIMPLE-REFLEX-AGENT(*percept*) **returns** an action
  persistent: *rules*, a set of condition–action rules

  state ← INTERPRET-INPUT(*percept*)
  rule ← RULE-MATCH(*state*, *rules*)
  action ← *rule*.ACTION
  return *action*

♣ Limited intelligence

It will work **only if** the correct decision can be made based on only the current percept, i.e., **only if** the environment is fully observable.

Model-based reflex agent

Model the state of the world by modeling how the world changes & how its actions change the world

In partially observable task environments

- Need to keep track of unseen portions of the world

Memory

- Internal state: guess of unobserved portions

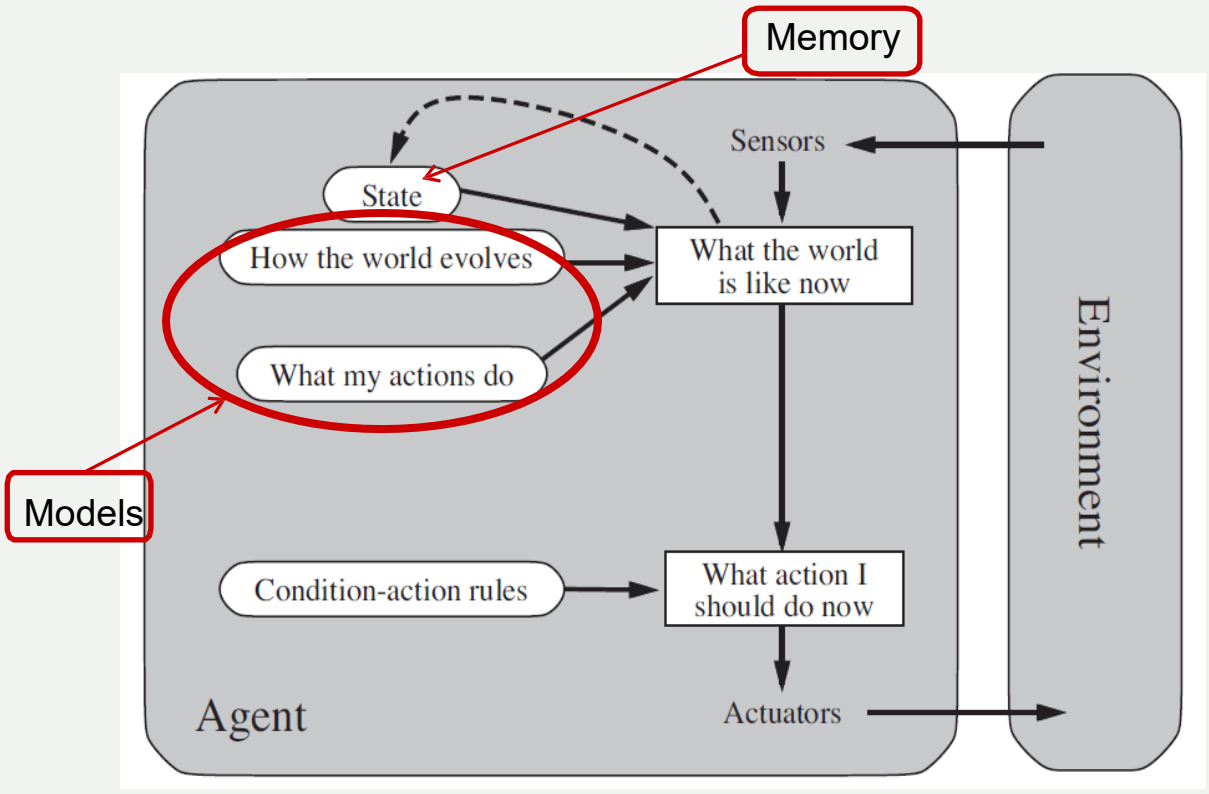
Models

- How the world evolves
- What my actions do

Examples

- searching a maze
- passing a car in a single-lane road

Model-based reflex agent



- Partially observable environment.
- Need to maintain some internal state.
- Update it using *knowledge*.

⇩

Model of the world

  - How does the world change?
  - How do actions affect the world?

Model-based reflex agent

```
function MODEL-BASED-REFLEX-AGENT(percept) returns an action
  persistent: state, the agent's current conception of the world state
              transition_model, a description of how the next state depends on
                    the current state and action
              sensor_model, a description of how the current world state is reflected
                    in the agent's percepts
              rules, a set of condition–action rules
              action, the most recent action, initially none

  state ← UPDATE-STATE(state, action, percept, transition_model, sensor_model)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
```

- It is rarely possible to describe the exact current state of the environment.
- The maintained "state" does not have to describe the world.

Goal-based agent

Goals provide reason to prefer one action over the other.
We need to predict the future: we need to plan & search

Best current action is chosen by higher-level goals, not just the current percept

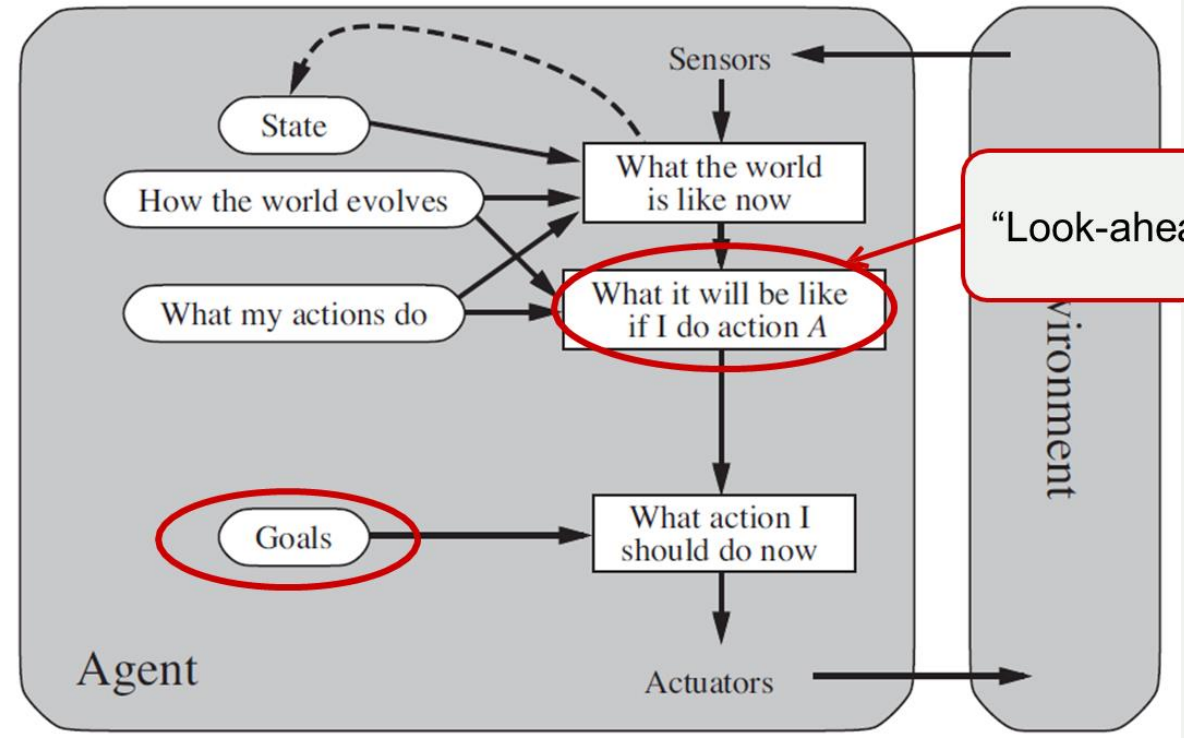
Goal: “arriving at destination x” for automated taxi

Find a sequence of actions to achieve the goal

Actions: “left turn, drive 0.5 mile, right turn, drive 1.2 miles,…”

Main design for Part 1 of the course

Goal-based agent

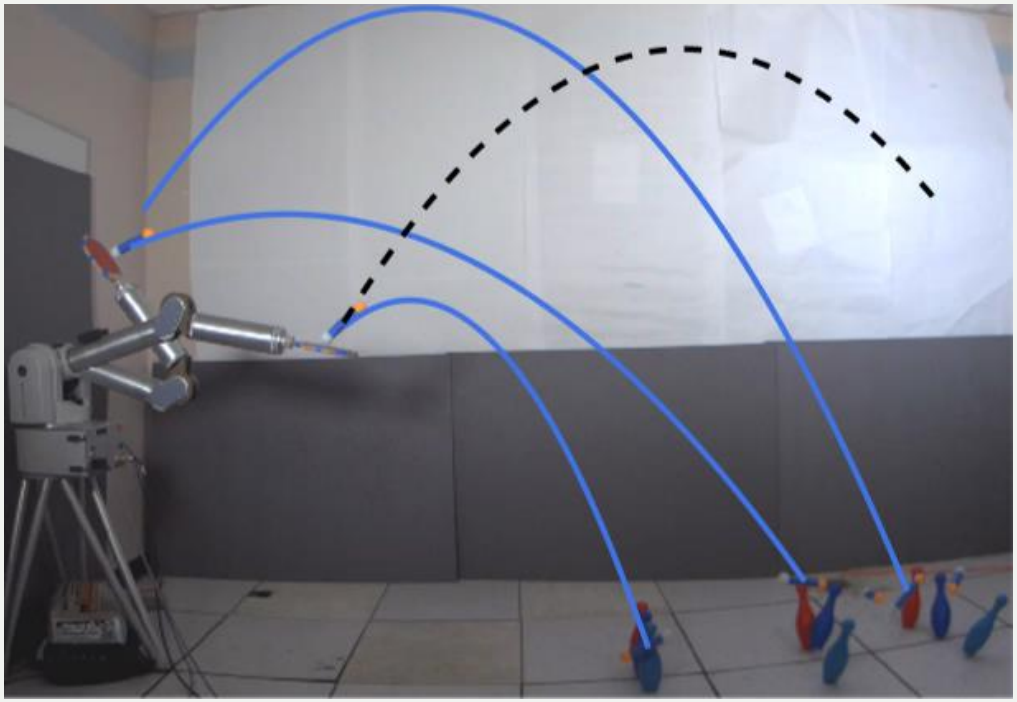


- Needs also some goal information describing desirable situations.
- *Search* and *planning*

  when a long sequence of actions is required to find the goal.
- Difference in *taking the future into account*.

Goal-based agent

Goal Hit an in-flight object to a target.



https://www.youtube.com/watch?v=dGBevZ54E3s

IEEE Transactions on Robotics, vol. 38, no. 5, pp. 3187-3202, 2022

- Sensors: 2 high-speed cameras
- Actuators: 4-DOF WAM Arm
- Models: impact dynamics, aerodynamics & robot kinematics
- State estimation: uses an extended Kalman filter (EKF) to track the velocity & angular velocity of the flying object.
- Planner: determines the time instant of hitting and the bat's position and velocity at this instant.

Utility-based agent

Some solutions to goal states are better than others.
Which one is best is given by a utility function.
Which combination of goals is preferred?

Goal-based agent: successful or not

Multiple goals: taxi

- Goal 1: arrive at destination
- Goal 2: minimize time/fuel consumption
- Goal 3: passenger safety/comfort

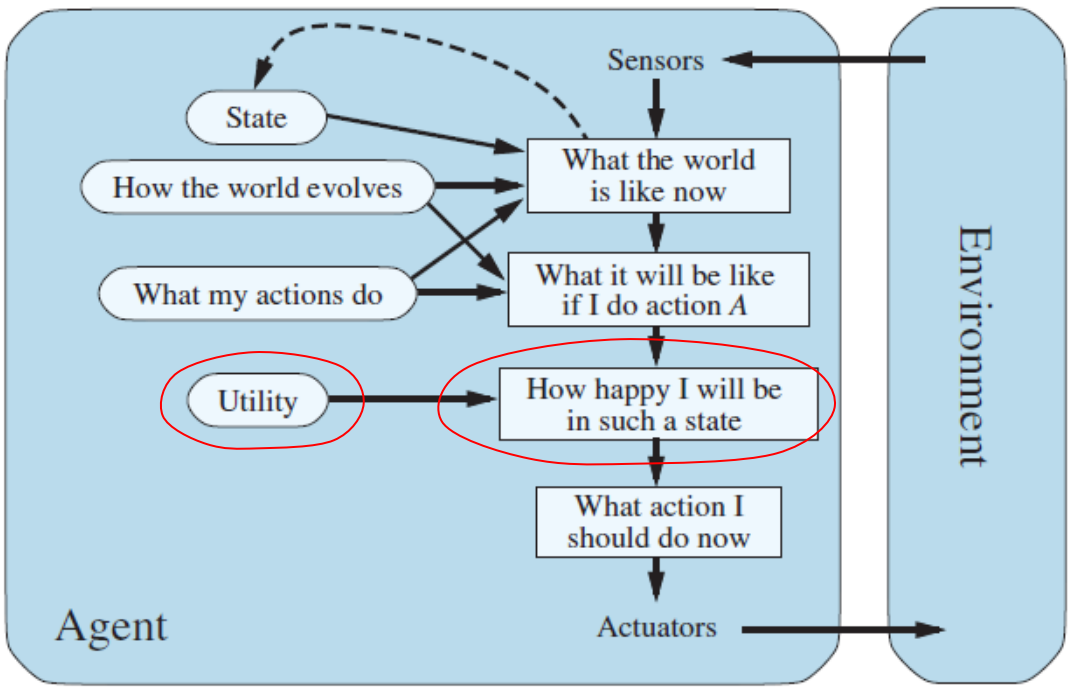
Utility can resolve multiple conflicting goals

- Maximize the utility as the weighted sum of multiple goals

Furthermore, if the environment task is stochastic, maximize the *expected* utility

More in depth in Part 3 Utility theory

Utility-based agent



- Different ways to achieve a goal sometimes.
- Use a *utility function* that maps a (sequence of states) to a real number (*utility*)

  ↑

  internal performance measure
- Maximize expected utility.

- Goal improvements:
  - selection among conflicting goals
  - selection based on likelihood of success and goal importance

Learning agent

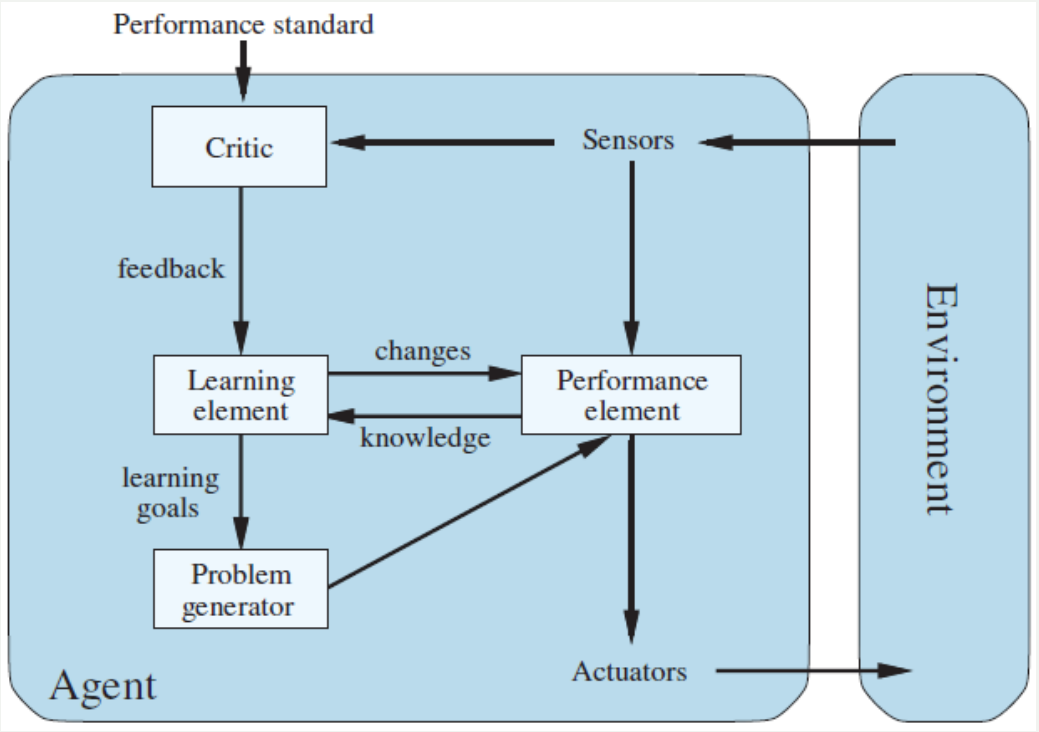
If we have intermediate feedbacks (in addition to final evaluation) on the agent's performance, use it to refine the agent function

Components of a learning agent

- Performance element (=non-learning part)
- Critic
- Learning element
- Problem generator

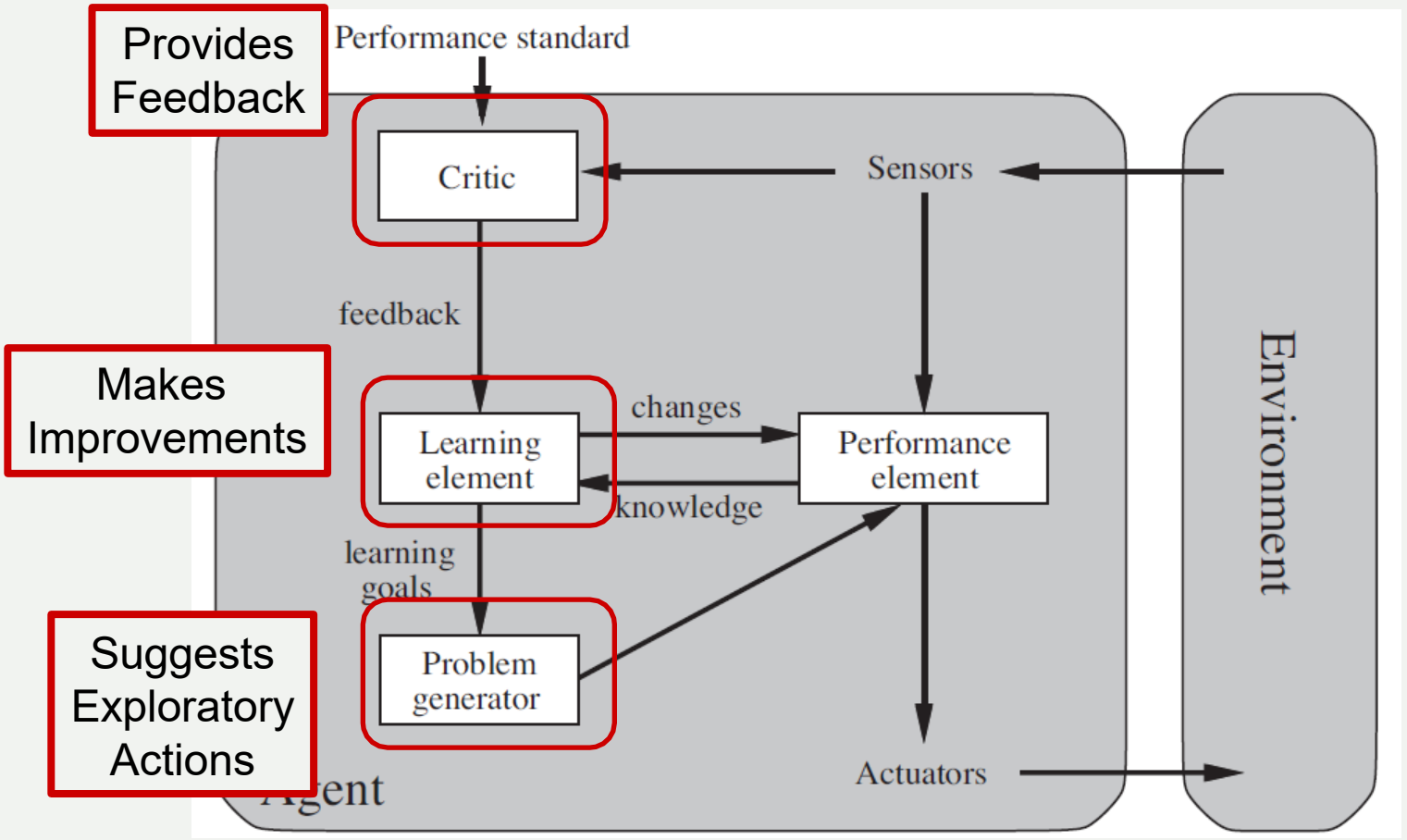
More in depth in Part 3 machine learning

Learning agent



- *Critic* provides feedback on the agent's performance based on fixed performance standard.
- *Learning element* Introduces improvements in performance element.
- *Performance element* selects actions based on the percepts.
- *Problem generator* suggests actions that will lead to new and informative experiences.

Learning agent



Reminder

Reading (Ch 2)

Quiz 1



- Q1 : True or False? (3pts)
  - a) The task environment of a taxi-driving agent is episodic
  - b) A rational will always achieve its goal
- Q2. Consider two intelligent agents playing chess with a clock. One of them is called “Deep Blue”, while the other is called Gary Kasparov (7pts)
  - a) Roughly specify the task environment for Deep Blue. (This means specify each letter in “PEAS”)
  - b) Determine each of the following properties of this task environment: 1) fully observable or partially observable, 2) deterministic or stochastic, 3) episodic or sequential, 4) static or dynamic or semidynamic, 5) discrete or continuous, 6) single agent or multi-agent. Explain you answer

1.(2pts) **Chess Agents** Consider two intelligent agents playing chess with a clock. One of them is called "Deep Blue", while the other is called Gary Kasparov.

a.(1pt) Roughly specify the task environment for Deep Blue. (This means specify each letter in "PEAS").

a) answer: a) performance measure: winning the game, environment: chess-pieces on a chess-board, adversary, c) actuators: screen, d) camera, keyboard.

b.(1pt) Determine each of the following properties of this task environment: a) fully observable or partially observable, b) deterministic or stochastic, c) episodic or sequential, d) static, dynamic, or semidynamic e) discrete or continuous, f) single agent or multi-agent. Explain your answer.

b) answer: a) fully observable, b) deterministic or strategic are both ok. c) sequential, d) semidynamic, e) discrete f) multi-agent.

- Rational ≠ omniscient or clairvoyant
  - Percepts may not supply all relevant information
  - Action outcomes may not be as expected
  - Hence, rational ≠ successful (e.g., lottery)

taxi driving	partial	stochastic	sequential	dynamic	continuous	multi